TWO QUESTIONS FOR THE ROOM

Your AI remembers.

But is it right?

Have you ever corrected an AI—then seen it use the old fact again?

What if that AI could act on the mistake?

TraceWise AI • Memory assurance for AI agents, assistants and workflows

Confident AI answers can have real consequences

Three documented failures — a liability precedent, a priced-in loss, and a system left running.

LEGAL PRECEDENT • AIR CANADA, 2024

Liable

The airline argued its chatbot was a "separate legal entity" answering for itself. The tribunal called that remarkable and rejected it.

What your assistant says is what your company said.

Moffatt v. Air Canada, 2024 BCCRT 149

PRICED-IN LOSS • ZILLOW, 2021

$304M

Written off its home inventory in one quarter, after the pricing model's view of the market fell behind the market. Zillow guided to a further $240–265M and wound the business down.

A stale view of the world, priced at scale.

Zillow Group Q3 2021, Form 8-K • ~25% of staff cut

NO GATE • NYC MyCITY, 2024

Left live

A city chatbot told business owners they could take workers' tips and refuse tenants on rental assistance. After reporting, it stayed online.

No test existed that would have caught it.

The Markup / THE CITY, April 2024

Three different causes — a fabricated policy, a lagging model, an untested release. One shared gap: nothing tested the answer first.

One memory failure. Many different consequences.

Illustrative scenarios — wherever yesterday's context drives today's answer or action.

PERSONAL AI

Travel dates changed

Assistant plans around the old dates

WORKPLACE AGENT

Candidate availability updated

Scheduling agent picks the old time

ENGINEERING AI

Safety limit added last week

Copilot answers as if it never existed

SERVICE CHATBOT

Refund policy shortened

Customer receives outdated guidance

Two directions, one gap: STALE FACT KEPT · CURRENT FACT LOST

TraceWise: a memory check for AI systems

A repeatable assurance test for agents, assistants and AI-powered workflows.

01 REPLAY

Give the system dated facts and later updates

Python · PyYAML
dated session fixtures

02 PROBE

Ask targeted test questions

LiteLLM gateway
Claude Opus · disk cache

03 VERIFY

Compare with an independent answer key

Deterministic scorer
pytest regression suite

04 EXPLAIN

Flag errors and show supporting evidence

Streamlit dashboard
JSONL run log

Memory backends under test: **full history** · **LangGraph-style latest-per-key reducer** · **LiteLLM token-budget trim** Service: **FastAPI**

Every backend implements one frozen interface, so the memory layer is the only variable. TraceWise tests accessible, controlled memory systems; it cannot read hidden third-party AI memory.

NOW: WALK THROUGH THE PROTOTYPE

Let’s test what the AI remembers.

1. Old + new information
2. What changed, and who missed it
3. One probe, all systems
4. Open a failure, see its evidence

DEMO • SWITCH TO THE BROWSER APP

SOURCE REPOSITORY ↗

Demo uses synthetic client-preference changes; the same method applies to other authorised AI workflows.

Before: you trust it. After: you can test it.

The goal is a clear engineering decision, not just a score on a chart.

WITHOUT A MEMORY CHECK

“Yes — that suits your crypto-heavy profile.”

Sounds right, cites the file
No systematic update test
Failure discovered late

WITH TRACEWISE

FLAG → EVIDENCE → RETEST

Wrong answer is identified
Old and applicable facts shown
Owner can investigate the cause

Now: measurable failures and comparison • Later: suggested fixes and optional release gates

What we measured: the memory layer moves accuracy 64 points

100 dated questions · 3 memory layers · 300 live answers · Claude Opus, one system prompt, scored by code.

MEMORY LAYER	CORRECT	COULDN'T ANSWER	TOKENS / QUERY
Keep everything	**88** /100	0	1,086
Latest-per-key reducer	**86** /100	5	480
Trim to token budget	**24** /100	49	250

The engineering decision: **44% of the tokens for 2 points of accuracy** — and the lowest cost per correct answer, $0.0044 against $0.0074. Trimming is cheapest per query and dearest per correct answer: $0.0207, 4.7× the reducer.

THE RESULT WE DIDN'T WANT

We expected stale facts to poison answers. So we deleted the one line telling the model to prefer recent facts, and re-ran all 300.

12 answers broke

None of them had a stale fact in context — though 58 other answers did. Every loss landed on the layer that **drops** facts. It fails by forgetting.

Logs: results/run_log.opus.jsonl and .recency-off.jsonl · rerun with scripts/analyse_ab.py · abstentions count as incorrect · rates confirmed 2026-10-25 in config/pricing.yaml.

Who uses TraceWise — and who benefits?

Main track: Governance & Compliance • Supporting: Explainability & Transparency

AI ENGINEER

Test memory updates in assistants, retrieval apps and agents

ASSURANCE TEAM

Keep failure evidence and review high-risk changes before rollout

INDIVIDUAL · NOT YET

Eventually test an authorised personal agent or workflow

Test → Record evidence → Flag risk → Human review

Personal integration is future work. The prototype is an assurance tool, not a safety or compliance certification.

One assurance engine. Different ways to use it.

From a controlled hackathon benchmark to integrations for teams and individuals.

TODAY

Browser prototype

Synthetic histories, target questions, actual answers and evidence

NEXT

Developer integration

Python package / API / CI checks for team-owned AI workflows

FUTURE

Personal agent check

Opt-in history import and checks for individual AI assistants

Test AI memory before decisions depend on it.

References and scope

Every figure on the evidence slide, with the document it came from.

Moffatt v. Air Canada, 2024 BCCRT 149

Chatbot gave incorrect bereavement-fare advice. The airline's "separate legal entity" argument was rejected; CAD $650.88 awarded. The cause was never established as a memory failure.

Zillow Group Q3 2021 results — Form 8-K, Exhibit 99.1

~$304M inventory write-down in the Homes segment, $240–265M more guided for Q4, ~25% of staff cut. A pricing-model failure, not a memory failure.

The Markup / THE CITY — NYC MyCity chatbot (2 April 2024)

City chatbot advised business owners they could take workers' tips and refuse tenants on rental assistance. It remained publicly accessible after publication.

EU AI Act — Article 12

Record-keeping obligations for qualifying high-risk AI. TraceWise produces evidence logs; it confers no automatic compliance.

TraceWise repository — logs, answer key and benchmark

Synthetic histories, three memory backends, 300 committed live answers per arm, and the deterministic scorer that produced slide 07.

Slide 03 scenarios are illustrative. TraceWise cannot audit hidden memory inside external AI services.